

SWIMMING

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Swimming Australia Uses Data and Machine Learning on AWS to enhance Athlete Performance

Watch their story

"I want swimming to be the Formula 1 of Olympic Sport. Our work with AWS, especially in data and analytics, has given us the confidence and edge to push boundaries, and we're not done yet. There's always something more to learn, more to improve."

Jessica Corones
*General Manager of Performance
Australia*



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Swimming Australia, the national governing body for competitive swimming, is at the heart of a sport that is part of Australia's DNA. With over 100,000 registered members across 1,100 clubs nationwide, the organization has built a legacy of creating world champions and driving Australia's Olympic success—amassing 239 medals, including 78 golds. Today, it remains committed to pushing the boundaries of athletic performance.

To maintain its legacy on the world stage, Swimming Australia sought to transform the vast amounts of raw performance data it had on its athletes into actionable, real-time insights. Working with [Amazon Web Services](#) (AWS), the organization harnessed Machine Learning and real-time analytics to craft winning strategies and help its athletes swim faster, and further.

Overcoming Data Challenges to Unlock Athletic Potential

Swimming Australia envisioned a future where athlete data could drive more strategic coaching and unlock new levels of performance. However, critical metrics—such as race analysis, training splits, and medical reports—were scattered across different platforms and devices. This fragmentation made it challenging for coaches to gain a comprehensive view of each athlete's progress.

Jessica Coronas, general manager of Performance Support and Olympic Campaign at Swimming Australia, explained, **“We knew we had the data and we knew there was more opportunity in it if we could harness it and understand it, but it was siloed. It was sitting on laptops, it was all over the place, and so to bring it all together into the data lake so that we were able to connect the data sets was critical.”**





Another pressing issue was the need for real-time feedback during training. Coaches relied on manual video annotations and personal observations which, while effective, were time-consuming and lacked the immediacy needed to correct techniques during training. These limitations were felt acutely in the relay team selection process where timely, data-driven decisions could mean the difference between a podium finish and a missed opportunity.



Building Specialized Training Tools Using AWS-Powered Insights

To drive strategic coaching and deliver real-time feedback, Swimming Australia used AWS to develop two new coaching tools: Lane 4 and the Relay App.



Lane 4 is a web portal that centralizes performance data, making it easily accessible across devices. Coaches can retrieve real-time insights, such as stroke rates, split times, and video analysis, allowing them to provide immediate, actionable feedback to athletes during training sessions.

Meanwhile, the Relay App focuses on optimizing relay team performances. It allows coaches to simulate various team lineups and predict race outcomes, providing a tactical edge in high-pressure competitions. By factoring in critical elements like changeover times and team dynamics, the app helps coaches make more informed decisions on team order and strategy. The app was especially beneficial during the Tokyo 2020 Olympics, where Australia secured medals in all six relay events.

With its data in one consolidated location, Swimming Australia could then utilize [Amazon SageMaker](#) to automate race tracking and competitor strategy analysis. Using [Amazon QuickSight](#)'s interactive dashboards, coaches could visualize data easily, which facilitated immediate technique review and informed decision-making.

“With AWS’s support, we’ve been able to harness our data in real-time, allowing our coaches to make better, faster decisions on the pool deck. It’s given us a competitive edge, providing immediate feedback and helping us optimize performance throughout the training process,” said Corones.

Advancing Real-Time Analysis for Future Competitive Edge

Swimming Australia aspires for swimming to be the "Formula 1 of Olympic sports", continually pushing the boundaries of athletic performance. By refining tools like Lane 4 and expanding real-time analysis capabilities, the organization plans to deliver increasingly personalized feedback to its athletes, ensuring they remain competitive on the global stage.

As Swimming Australia gears up for future Olympic games, including Brisbane 2032, it is committed to advancing training environments with cutting-edge AI and machine learning. Through AWS, Swimming Australia continues to explore innovative ways to harness data. This approach elevates its training methods and supports athletes in their pursuit of excellence.



“The pursuit to be faster will never end. Even when you think you've seen the greatest swimmer of all time, along comes another one. Who knows how fast they can swim? That's the fun of it—you just never know,” said Corones. **“Our work with AWS, especially in leveraging data, has given us the confidence and edge to push boundaries, and we’re not done yet. There’s always something more to learn, more to improve.”**

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